Ch 5.1.3-4: k-Fold Cross-Validation

Lecture 11 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Wed, Oct 5, 2022

Announcements

Last time:

- Validation Set
- LOOCV

Announcements:

- Posted HW #4 – due Monday
- 5.1

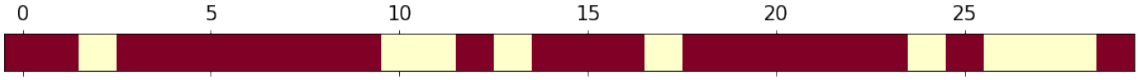
Covered in this lecture

- k-fold CV
- ~~CV for classification~~

Section 1

Last time

Validation set approach

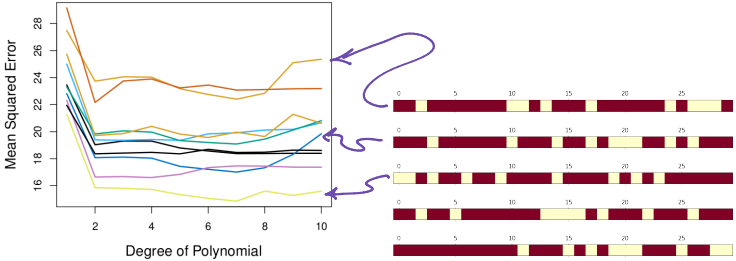


- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

data pt indices

not input variables

Problems



Ex. Predict `mpg` using `horsepower`

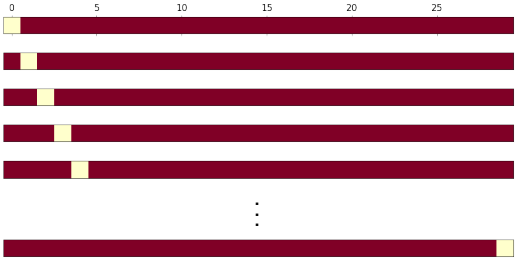
- Highly variable results, no consensus about the error
- Tends to overestimate test error rate

Leave One Out CV (LOOCV)

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

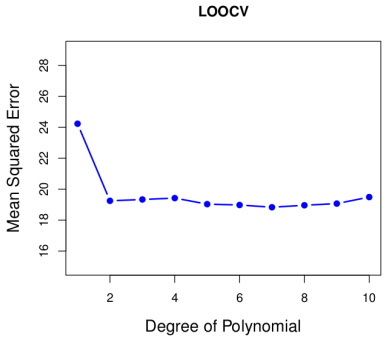
- Rinse and repeat



Return the score:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{MSE}_i$$

Pros and Cons



- No variance
- Higher computation cost

Speeding up LOOCV

Warning: This only works for least squares linear or polynomial regression.

$$h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{j=1}^{n}(x_j - \overline{x})^2}$$

leverage

$$\frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i = CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

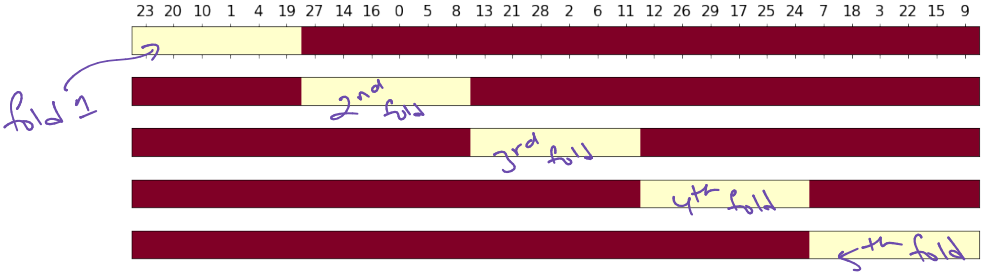
Section 2

k-Fold CV

The idea

$k = 5$ $n = 30$

Random shuffle of indices $\mathcal{R}$



every data point is in exactly one fold

Mathy version

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} (y_j - \hat{y}_j)^2$

  # pts in U_i

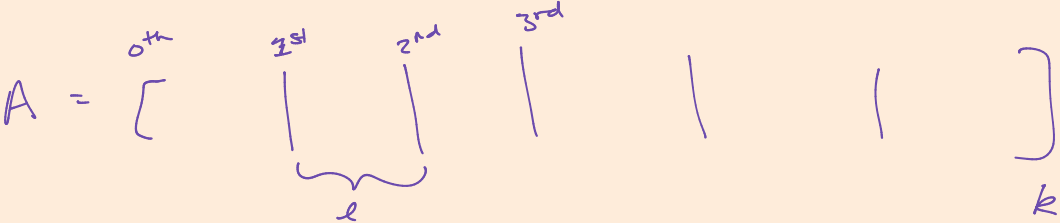
  data in fold (test)
- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{MSE}_i$$

k folds

Coding - Building k-fold CV



$A[\ell \cdot i : \ell \cdot (i+1)]$

for i in range(k):

Pros and Cons

Pros:

Cheaper computation

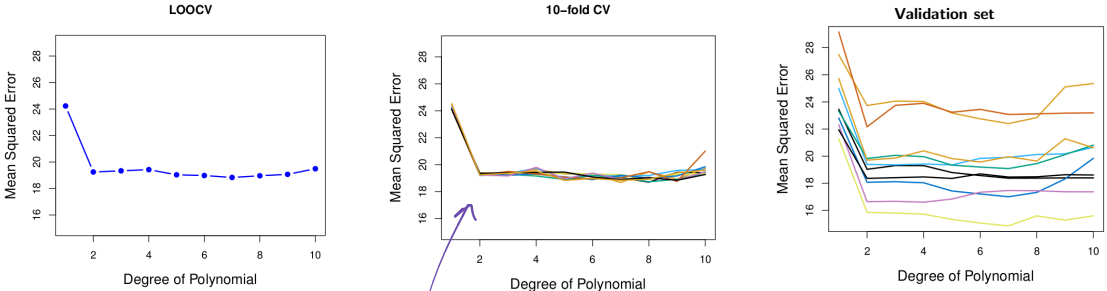
$k = 5, 10$ are reasonable choices

Cons:

- Some variability (vs LOOCV had no variability)
- Lower variability than validation set approach

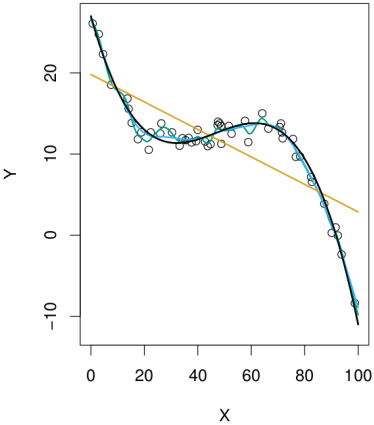
Comparison

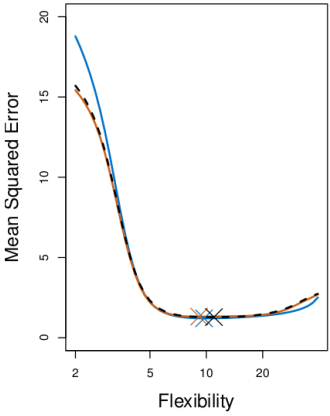
Start here next time



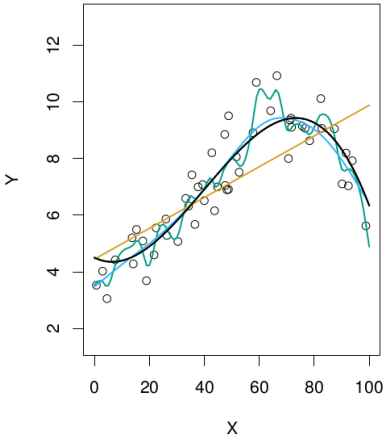
different shuffle of my data to start

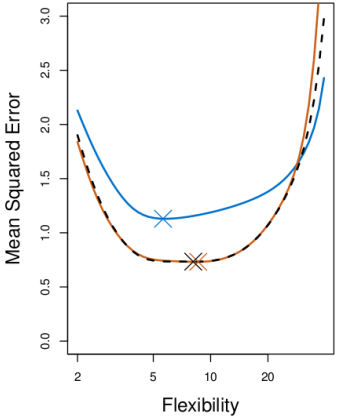
Comparison with simulated data: Ex 3



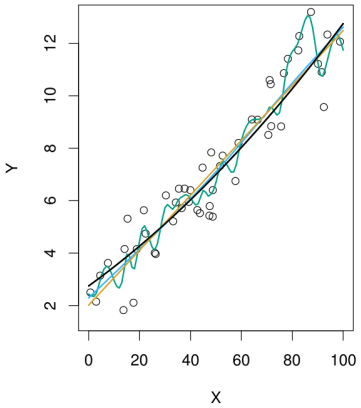


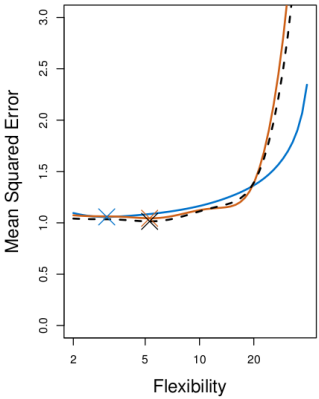
Comparison with simulated data: Ex 1





Comparison with simulated data: Ex 2





Takeaways from the examples

Bias-Variance Tradeoff: Bias

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Bias-Variance Tradeoff: Variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Do the remainder of the coding

Next time

14	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
15	W	Oct 5	k-fold CV	5.1.3	
16	F	Oct 7	More k-fold CV	5.1.4	
17	M	Oct 10	CV for classification	5.1.5	HW #4 Due
18	W	Oct 12	Resampling methods: Bootstrap	5.2	
19	F	Oct 14	Subset selection	6.1	
20	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
21	W	Oct 19	Shrinkage: Lasso	6.2.2	
22	F	Oct 21	Dimension Reduction	6.3	
	M	Oct 24	No class - Fall break		
21	W	Oct 26	More dimension reduction; High dimensions	6.4	
22	F	Oct 28	Polynomial & Step Functions.	7.1,7.2	HW #6 Due
23	M	Oct 31	Review		
24	W	Nov 2	***Midterm #2***		